

## Concepts & Methods in Law & Economics

### Problem set 2: Primer on Game Theory

Prof. Dr. Roee Sarel

(Q1) Two regulatory bodies who are in charge of the same market are faced with the following payoff structure:

|                | Regulate | Don't regulate |
|----------------|----------|----------------|
| Regulate       | 1, 1     | 1, 0           |
| Don't regulate | 0, 1     | 0, 0           |

Is there a dominant strategy in this game? If so, is it a “strictly dominant strategy” (=gives the highest payoffs in all states of the world) or “weakly dominant strategy” (=gives the same payoff, or the highest payoff, in all states of the world)?

Let us start with player 1, who chooses a row (player 2 thus chooses a column). Suppose player 1 chose to regulate – player 2 then either also regulated (getting 1) or does not regulate (getting 0). As  $1 > 0$ , player 2 regulates. The game is symmetrical so we can deduce the same for player 1. Hence: (regulate, regulate) is a nash equilibrium. But is it a dominant strategy? Suppose player 1 does not regulate – player 2 then either chooses to regulate (getting 1) or not regulate (getting 0), so again – player 2 prefers to regulate. The same can be said for player 1. Thus, to regulate is also a dominant strategy – it is the best response to any action taken by the other player. This dominant strategy is a **strictly** dominant strategy, because it always yield 1 which is strictly larger than the alternative (which is 0).

(Q2) Find the nash equilibrium (or multiple nash equilibria, if there is more than one) in the following game:

|    | A1  | A2   | A3   |
|----|-----|------|------|
| B1 | 5,5 | 7,6  | 9,5  |
| B2 | 4,7 | 0, 0 | 2,3  |
| B3 | 7,4 | 3,3  | 15,2 |

There are 2 Nash Equilibria (“NE”) in this game: (B1, A2) and (B3, A1). To illustrate: if player 1 chooses “B1” then player 2’s best response is to play “A2” because it yields 6 and the other options yield 5. Respectively, If player 2 plays “A2” then player 1’s best

response is A2 because it yields 7 and the other options yield less (0 or 3). Following a similar logic, if player 1 chooses B3, player 2's best response is A1 and vice versa.

### (Q3)

Find all pure strategy Nash-Equilibria of the games a.) to d.) and the subgame perfect equilibria of e.) to h.).

a.)

|          |   | Player 2 |     |
|----------|---|----------|-----|
|          |   | L        | R   |
| Player 1 | U | 2,3      | 3,2 |
|          | D | 1,2      | 2,1 |

b.)

|          |   | Player 2 |     |
|----------|---|----------|-----|
|          |   | L        | R   |
| Player 1 | U | 3,3      | 8,2 |
|          | D | 2,8      | 6,6 |

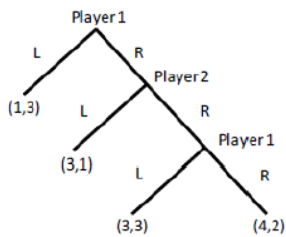
c.)

|          |   | Player 2 |     |
|----------|---|----------|-----|
|          |   | L        | R   |
| Player 1 | U | -1,3     | 3,2 |
|          | D | 2,1      | 2,3 |

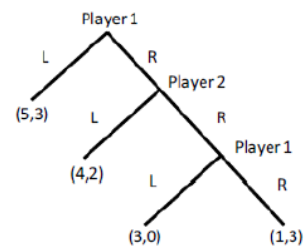
d.)

|          |   | Player 2 |     |     |
|----------|---|----------|-----|-----|
|          |   | L        | M   | R   |
| Player 1 | U | 2,3      | 1,2 | 1,2 |
|          | M | 3,2      | 2,3 | 4,1 |
|          | D | 4,3      | 1,2 | 3,1 |

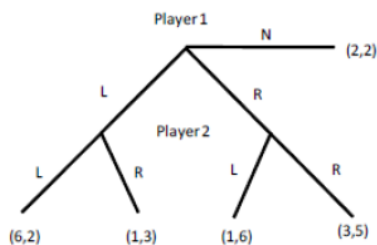
e.)



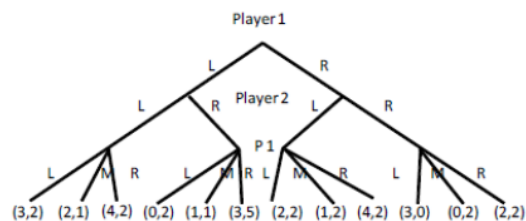
f.)



g.)



h.)



**Note: “Pure strategy” means that a player chooses one option. It is different from a “mixed strategy” where each player randomizes between options (we have not discussed that in class).**

**Answers – the NE are as follows:**

- a) (U, L). Note that player 1’s dominant strategy is “U” whereas player 2’s dominant strategy is “L”.
- b) (U,L). Note that this is a prisoner’s dilemma.
- c) This game has no NE in pure strategies.
- d) (L,D) and (M,M)

e) Solving by backward induction:

- In node three Player 1 would choose R (4 instead of 3)
  - Knowing this Player 2 would choose R in node 2 (2 instead of 1)
  - In node one Player 1 chooses R (4 instead of 1)
- => The outcome of the subgame perfect NE is (4; 2) ; with R,R,R.

- f) In node three Player 1 would choose L (3 instead of 1).  
Knowing this Player 2 would choose L in node 2 (2 instead of 0)  
In node one Player 1 chooses L (5 instead of 4)

=> Outcome of the subgame perfect NE is (5; 3) ; where player 1 just chooses L in node one.

- g) Player 2 knows in which node she is (i.e. she knows if Player 1 has chosen L or R)  
If Player 1 has chosen L Player 2 will choose R (3 instead of 2)  
If Player 1 has chosen R Player 2 will choose L (6 instead of 5)  
Player 1 chooses N (2 instead of 1)  
=> The outcome of the subgame perfect NE is (2; 2) ; where player 1 just chooses N in node one.

- h) The outcome of the subgame perfect NE is (4; 2) ; with actions along the equilibrium path of R,L,R.